ASSIGNMENT 1

```
#include <stdio.h>

int main()
{
   int household, previous, current, units;
   char category;

   printf("Enter Household Number: ");
   scanf("%d", &household);

   printf("Enter Category: ");
   scanf(" %c", &category);

   printf("Enter Previous Reading: ");
   scanf("%d", &previous);

   printf("Enter Current Reading: ");
   scanf("%d", &current);

   if(previous < 0 || current < 0 ||
     previous > 65535 || current > 65535)
      printf("Invalid Reading");

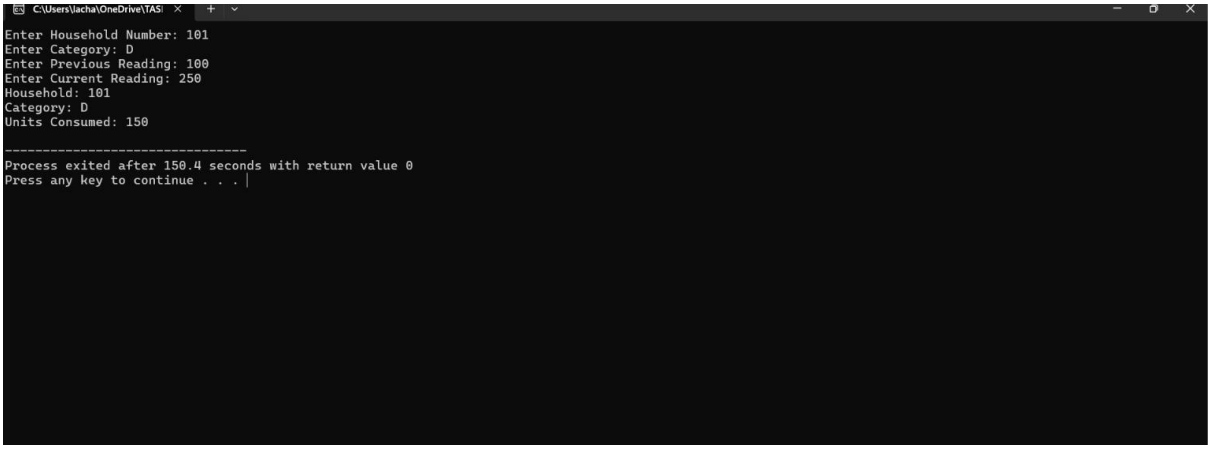
   else if(current >= previous)
      units = current - previous;
   else
      units = (65535 - previous) + current + 1;

   if(previous >= 0 && current >= 0 &&
     previous <= 65535 && current <= 65535)
```

```
    {
        printf("Household: %d\n", household);
        printf("Category: %c\n", category);
        printf("Units Consumed: %d\n", units);
    }

    return 0;
}
```

Output:



2.

```
#include <stdio.h>
#include <limits.h>
#include <float.h>

int main()
{
    int a;
    float b;
    double c;
```

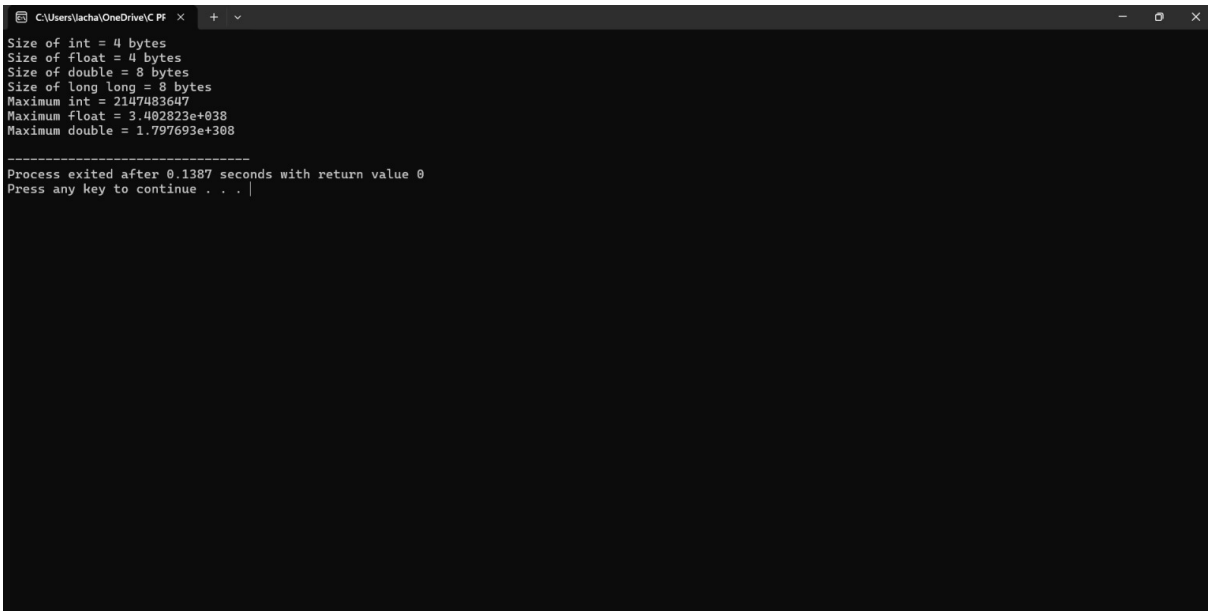
```
    long long d;

    printf("Size of int = %zu bytes\n", sizeof(a));
    printf("Size of float = %zu bytes\n", sizeof(b));
    printf("Size of double = %zu bytes\n", sizeof(c));
    printf("Size of long long = %zu bytes\n", sizeof(d));

    printf("Maximum int = %d\n", INT_MAX);
    printf("Maximum float = %e\n", FLT_MAX);
    printf("Maximum double = %e\n", DBL_MAX);

    return 0;
}
```

Output:



3.#include <stdio.h>

#define RATE1 2

#define RATE2 3

#define RATE3 5

```
#define FIXED 50

int main()
{
   int units, charge, bill;

   printf("Enter Units: ");
   scanf("%d", &units);

   if(units < 0)
      printf("Invalid Units");

   else if(units <= 100)
      charge = units * RATE1;

   else if(units <= 200)
      charge = 100 * RATE1 +
            (units - 100) * RATE2;

   else
      charge = 100 * RATE1 + 100 * RATE2 +
            (units - 200) * RATE3;

   if(units >= 0)
   {
      bill = charge + FIXED;

      printf("Energy Charge = Rs.%d\n", charge);
      printf("Total Bill = Rs.%d\n", bill);
   }
```

```
    return 0;
}
```

Output:



4.#include <stdio.h>

#define TO_PAISE(r) ((r) * 100)

#define SQUARE(x) ((x) * (x))

#define MAX(a,b) ((a) > (b) ? (a) : (b))

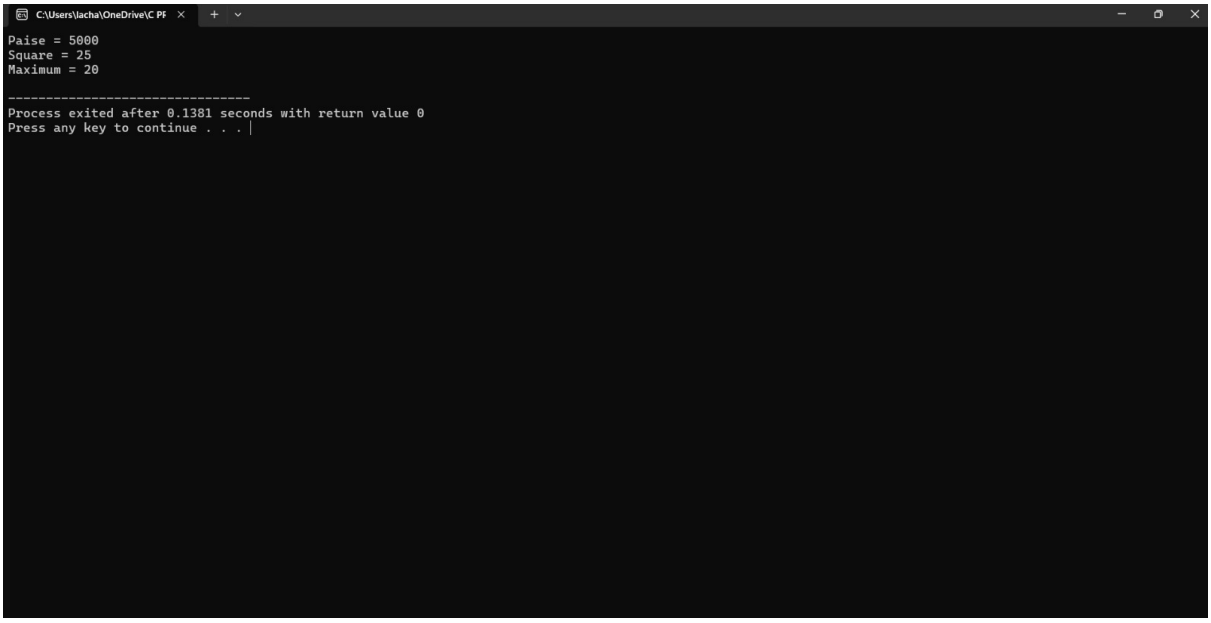
int main()

```
{
    int rupees = 50;
    int a = 10, b = 20;

    printf("Paise = %d\n", TO_PAISE(rupees));
    printf("Square = %d\n", SQUARE(5));
    printf("Maximum = %d\n", MAX(a, b));

    return 0;
}
```

Output:



5.

```
#ifndef BILLING_H
#define BILLING_H

#define RATE 2
#define FIXED 50

int calculate(int units);

#endif
```

```
#include <stdio.h>



int calculate(int units)
{
   return units * RATE + FIXED;
}

int main()
{
   int units, bill;

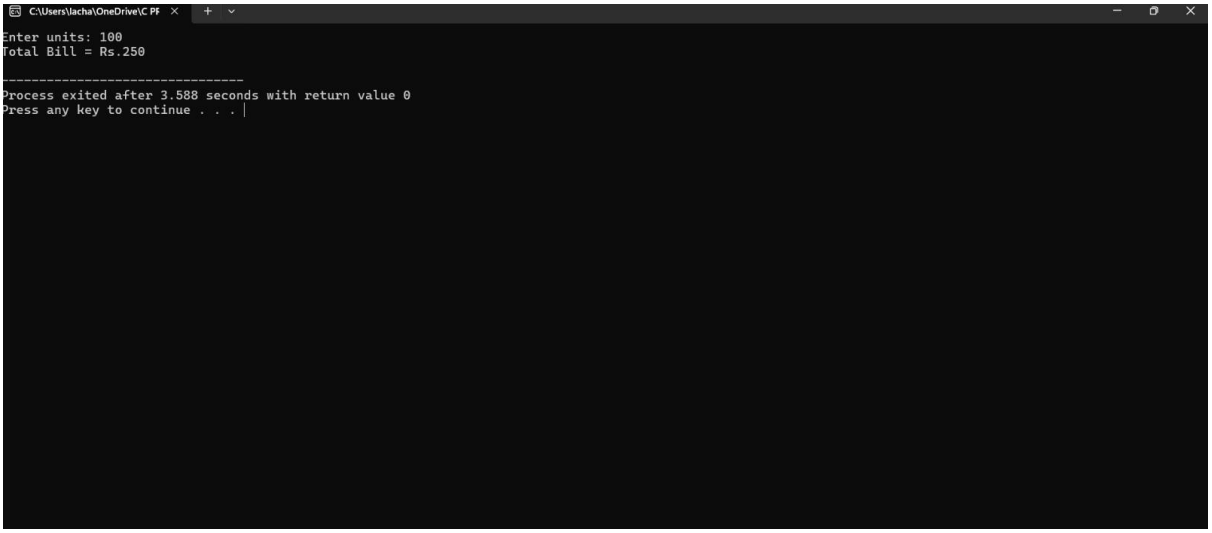
   printf("Enter units: ");
   scanf("%d", &units);

   bill = calculate(units);

   #ifdef DEBUG
   printf("Debug: Units = %d\n", units);
   #endif

   printf("Total Bill = Rs.%d\n", bill);

   return 0;
}
```
Output:



6.#include <stdio.h>

```
int main()
{
   int a, paise;
   float b;
   double c;
   char ch;
   char name[20];

   printf("Enter integer: ");
   scanf("%d", &a);

   printf("Enter float: ");
   scanf("%f", &b);

   printf("Enter double: ");
   scanf("%lf", &c);

   printf("Enter character: ");
   scanf(" %c", &ch);
```

```
    printf("Enter name: ");
    scanf("%19s", name);

    printf("Enter amount in paise: ");
    scanf("%d", &paise);

    printf("\nInteger = %d", a);
    printf("\nFloat = %.2f", b);
    printf("\nDouble = %.2lf", c);
    printf("\nCharacter = %c", ch);
    printf("\nName = %s", name);
    printf("\nAmount = Rs.%d.%02d",
           paise / 100, paise % 100);

    return 0;
}
```

Output:



7.#include <stdio.h>

```
int main()
{
    int a = 5, b = 2;
    float x;
    double bill = 162.50;
    long int value = 500000;
    int small;

    printf("Integer Division: %d\n", a / b);

    x = (float)a / b;
    printf("Floating Division: %.2f\n", x);

    printf("Implicit Conversion: %.2f\n", a + 2.5);

    printf("Explicit Conversion: %.2f\n", (float)a / b);

    printf("Bill Amount: %.2f\n", bill);

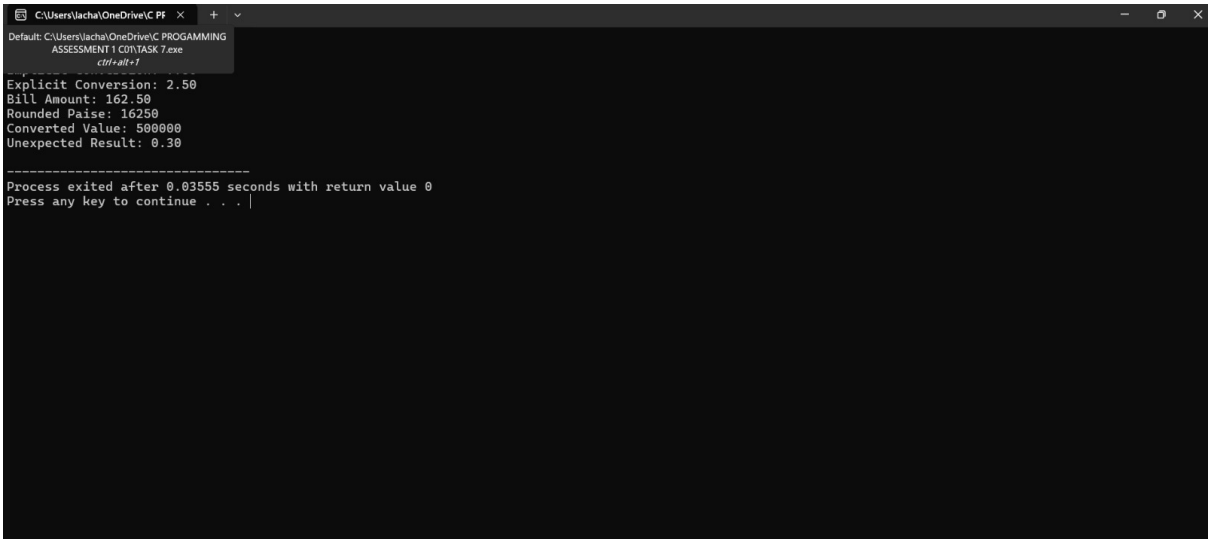
    printf("Rounded Paise: %lld\n",
          (long long)(bill * 100 + 0.5));

    small = (int)value;
    printf("Converted Value: %d\n", small);

    printf("Unexpected Result: %.2f\n", 0.1 + 0.2);

    return 0;
}
```

Output:



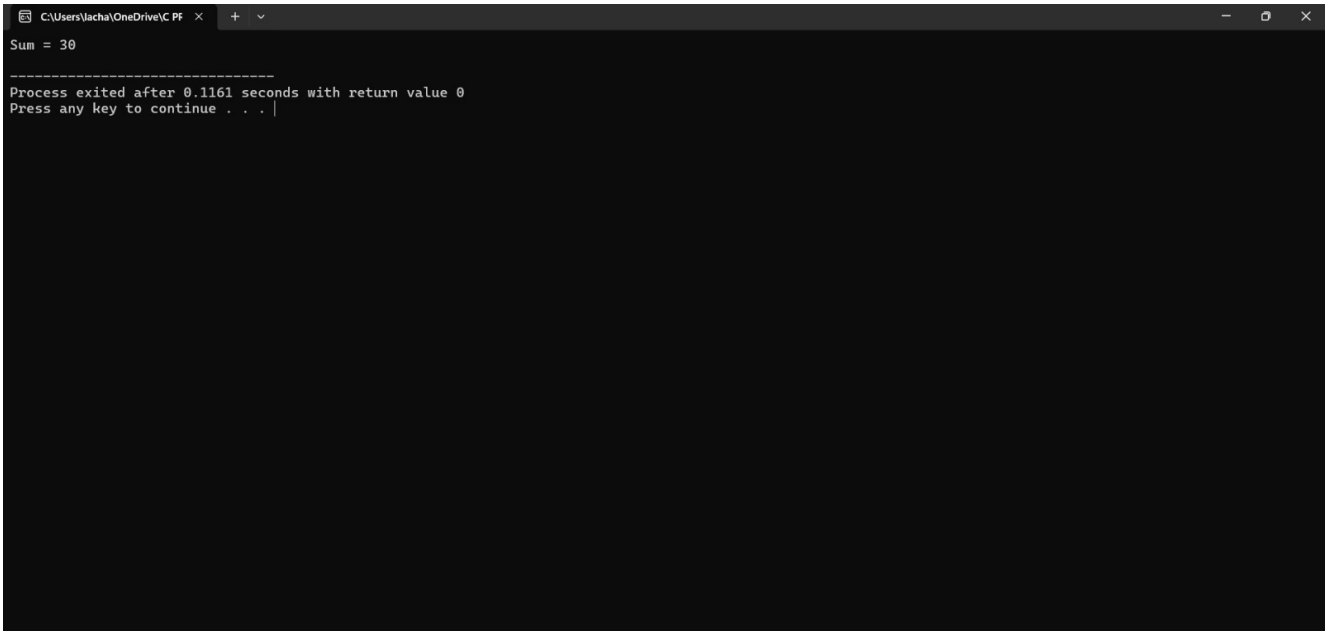
8.#include <stdio.h>

```
int main()
{
    int a = 10, b = 20;
    int sum = a + b;

    printf("Sum = %d\n", sum);

    return 0;
}
```

Output:



9.#include <stdio.h>

```
int main()
{
   int prev, curr, units;
   float bill;

   printf("Enter previous reading: ");
   scanf("%d", &prev);

   printf("Enter current reading: ");
   scanf("%d", &curr);

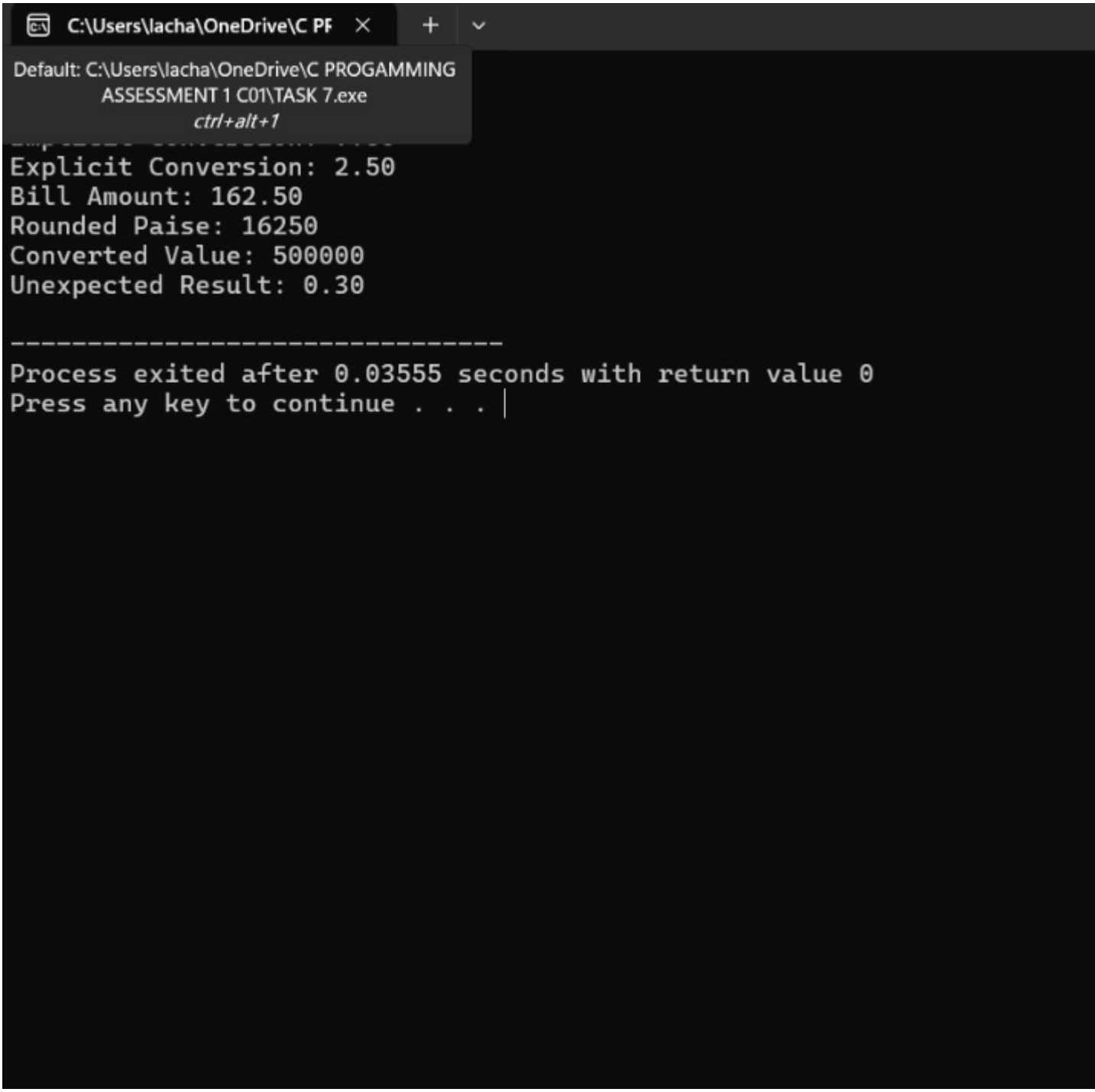
   if(prev < 0 || curr < 0 || prev > 65535 || curr > 65535)
      printf("Invalid reading");

   else
   {
```

```
        if(prev == 65535 && curr == 0)
            units = 1;
        else if(curr < prev)
        {
            printf("Invalid reading");
            return 0;
        }
        else
            units = curr - prev;

        if(units <= 100)
            bill = units * 1.5;
        else if(units <= 200)
            bill = 150 + (units - 100) * 2.5;
        else
            bill = 400 + (units - 200) * 4;
      printf("\nUnits = %d", units);
        printf("\nTotal Bill = Rs. %.2f", bill);
    }
 return 0;
}
```

Output:




```
#include <stdio.h>
#include <limits.h>

int main()
{
   int bills[] = {4000, 60000, 800000};
   int i;
```

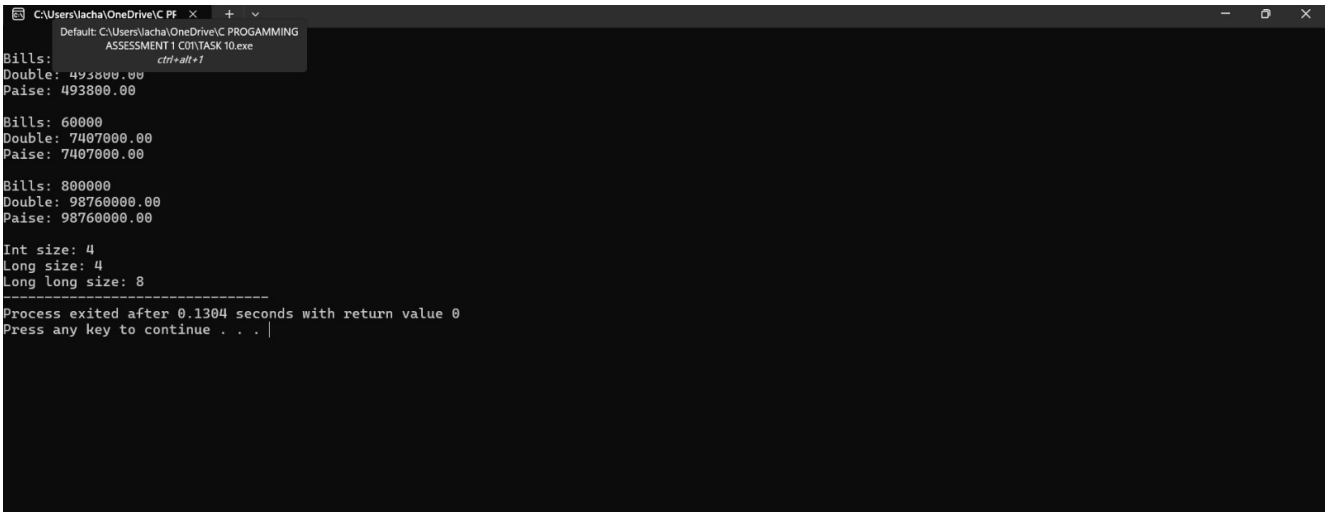
```
    for(i = 0; i < 3; i++)
    {
        double total = 123.45 * bills[i];
        long long paise = 12345LL * bills[i];

        printf("\nBills: %d", bills[i]);
        printf("\nDouble: %.2f", total);
        printf("\nPaise: %.2f\n", paise / 100.0);
    }

    printf("\nInt size: %zu", sizeof(int));
    printf("\nLong size: %zu", sizeof(long));
    printf("\nLong long size: %zu", sizeof(long long));

    return 0;
}
```

Output:



11.#include <stdio.h>

#define RATE1 1.5

```
#define RATE2 2.5
#define RATE3 4.0

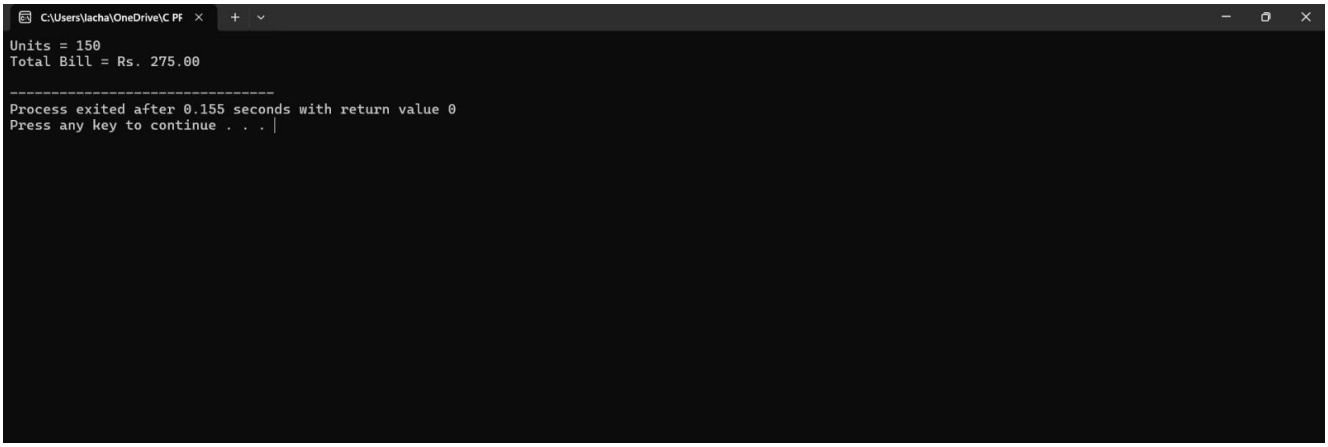
int main()
{
   int units = 150;
   double bill;

   if(units <= 100)
      bill = units * RATE1;
   else if(units <= 200)
      bill = 100 * RATE1 + (units - 100) * RATE2;
   else
      bill = 100 * RATE1 + 100 * RATE2
            + (units - 200) * RATE3;

   printf("Units = %d\n", units);
   printf("Total Bill = Rs. %.2f\n", bill);

   return 0;
}
```

Output:



12.#include <stdio.h>

```
int main()
{
    long long watt, total = 0;
    double rate, bill = 0;
    int hour, i;

    printf("Enter tariff rate: ");
    scanf("%lf", &rate);

    for(i = 1; i <= 5; i++)
    {
        printf("Enter watt-hours and hour: ");
        scanf("%lld %d", &watt, &hour);

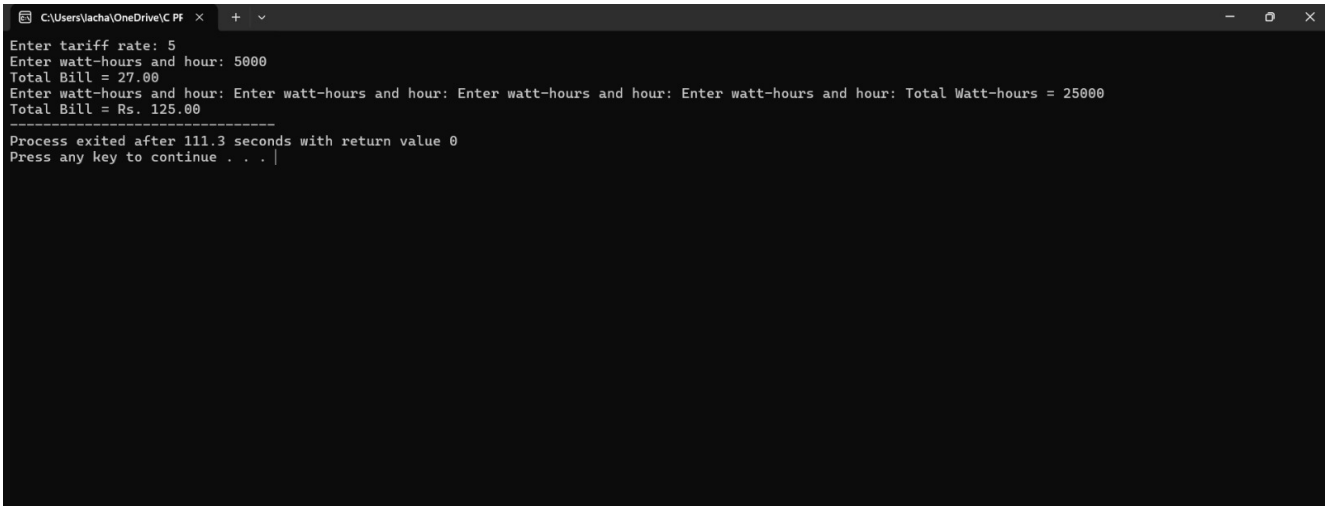
        total += watt;

        if(hour >= 18 && hour < 22)
            bill += (watt / 1000.0) * rate * 1.2;
        else
            bill += (watt / 1000.0) * rate;
    }

    printf("Total Watt-hours = %lld\n", total);
    printf("Total Bill = Rs. %.2f", bill);

    return 0;
}
```

Output:


C:\Users\lacha\OneDrive\C Pf
Enter tariff rate: 5
Enter watt-hours and hour: 5000
Total Bill = 27.00
Enter watt-hours and hour: Enter watt-hours and hour: Enter watt-hours and hour: Enter watt-hours and hour: Total Watt-hours = 25000
Total Bill = Rs. 125.00

Process exited after 111.3 seconds with return value 0
Press any key to continue . . .